

ACO-MoE-LoRA: Evolving-while-Training for Adapting Segment Anything Model 2 to Specialized Domains

Kaiyi Luo¹ Bangjun Wang¹ Li Zhang¹ Fanzhang Li¹ Fei Zhu¹ Jiaqing Fan¹

Abstract

Static fine-tuning paradigms impose rigid structural constraints on foundation models like the Segment Anything Model 2 (SAM2), limiting their adaptability to the varying complexity of specialized downstream tasks. To overcome this limitation, we propose **ACO-MoE-LoRA**, a dynamic framework that introduces an “Evolving-while-Training” strategy by synergizing Ant Colony Optimization (ACO) with a Latent Space Mixture-of-Experts (MoE) architecture. Central to our method is the ACO-ConvLoRA module, which employs a pheromone-guided routing mechanism to actively govern expert selection and topological evolution. By formulating expert assignment as an evolutionary pathfinding problem, this module effectively mitigates the standard routing collapse issue and enables elastic adjustment of LoRA ranks via weight slicing, bridging discrete structural search with continuous parameter training. Extensive experiments across 16 challenging datasets demonstrate that our framework consistently outperforms leading static adapters, while effectively addressing the local optimality limitations of recent dynamic heuristics. This work presents a self-organizing solution that harmonizes swarm intelligence with gradient optimization for efficiently adapting foundation models to specialized domains.

1. Introduction

The advent of foundation models, exemplified by the Segment Anything Model (SAM) series (Kirillov et al., 2023; Ravi et al., 2024; Carion et al., 2025), has revolutionized generic image segmentation. However, applying these

¹School of Computer Science and Technology, Soochow University, Suzhou, China. Correspondence to: Bangjun Wang <wangbangjun@suda.edu.cn>.

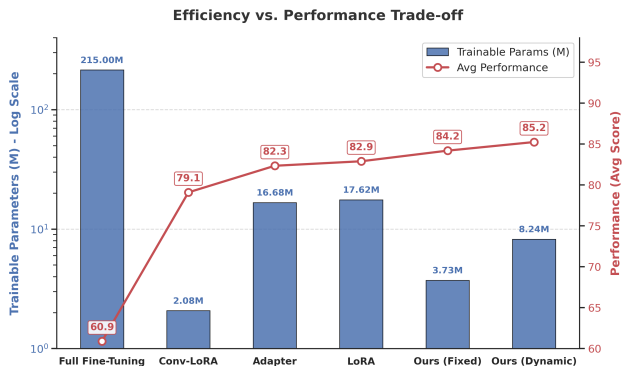


Figure 1. Efficiency-Performance Trade-off Analysis. This dual-axis chart compares trainable parameters (blue bars, log scale, left axis) against average segmentation performance (red line, right axis). The results highlight a distinct mismatch in *Full Fine-Tuning*, where significant parameter redundancy fails to translate into competitive performance. In contrast, our framework demonstrates a superior trade-off: “Ours (Fixed)” substantially reduces parameter overhead while maintaining high accuracy, and “Ours (Dynamic)” leverages evolutionary search to achieve the state-of-the-art performance.

general-purpose models to specialized domains (e.g., medical, marine) remains challenging due to significant domain shifts (Chen et al., 2023; Ma et al., 2024).

To adapt these models, Parameter-Efficient Fine-Tuning (PEFT) has become the standard. While static PEFT methods like LoRA (Hu et al., 2022) are efficient, they suffer from limited capacity due to fixed low-rank constraints. Recent works have attempted to mitigate this by introducing dynamic rank allocation (Zhang et al., 2023a; Valipour et al., 2023) or sparse adaptation (Ding et al., 2023), yet they often rely on gradient-based heuristics that struggle to escape local optima. To address this, Mixture-of-Experts (MoE) architectures have been proposed to scale capacity dynamically (Li et al., 2024; Wu et al., 2024). **However, directly applying standard MoE to foundation model fine-tuning faces two critical impediments:**

- **Representation Collapse & The Limits of Heuristic Balancing:** Traditional gradient-based gating networks are prone to “Routing Collapse” (Shazeer et al., 2017; Xue et al., 2024). While early approaches employed

auxiliary load-balancing losses (Fedus et al., 2022; Du et al., 2022) to encourage uniformity, the field is rapidly pivoting toward **Auxiliary-Loss-Free (ALF)** paradigms (Liu et al., 2024a; Wang et al., 2024) to mitigate gradient interference. However, these recent heuristics often balance load based solely on utilization counts, ignoring the semantic specialization of experts (Jing et al., 2025). This creates a need for a routing mechanism that is both gradient-free and performance-aware.

- **Parameter Explosion in Fine-tuning:** Standard MoE scales capacity by duplicating full neural blocks (e.g., FFNs or Convolutions) as experts (Riquelme et al., 2021; Fedus et al., 2022). In the context of fine-tuning large foundation models, maintaining a bank of full-weight experts acts counter to the goal of “parameter efficiency”, leading to prohibitive memory overhead and storage costs. Even recent PEFT-MoE hybrids (Wu et al., 2024; Li et al., 2024) typically duplicate the entire adapter module for each expert, which still results in linear parameter growth relative to the number of experts.

To reconcile the inherent tension between “High Capacity” (MoE) and “Parameter Efficiency” (PEFT), we propose a **Latent Space Expert** strategy. Departing from architectures that route inputs to independent adapters (Wu et al., 2024), we embed experts strictly within the low-rank bottleneck. This employs a “**Project-Route-Reconstruct**” workflow: features are projected into a compressed subspace via a shared matrix A , refined by lightweight experts, and expanded via B . This design yields a dual benefit:

- **Extreme Efficiency:** Experts operate on compressed latent vectors ($r \ll d$, e.g., 16 vs 256). This effectively decouples parameter cost from model width, reducing computational overhead by orders of magnitude.
- **Manifold Diversity:** Sharing the projection basis (A, B) ensures all experts operate within a unified semantic manifold, while the MoE layer enables diverse non-linear refinements in the latent space.

As shown in Figure 1, this strategy enables our Fixed Rank variant to surpass standard LoRA with extreme efficiency (3.73M params), while the Dynamic variant achieves state-of-the-art performance with minimal overhead.

To realize this transition from static to dynamic adaptation, we propose **ACO-MoE-LoRA**. This framework establishes an “Evolving-while-Training” paradigm by embedding Ant Colony Optimization directly into the training loop, actively reconciling topological flexibility with routing stability. Our core insight is to embed Swarm Intelligence—specifically

Ant Colony Optimization (ACO)—directly into the training loop as a dynamic meta-controller. This establishes an “Evolving-while-Training” paradigm that breaks the rigidity of fixed topologies. To realize this, we design the **ACO-ConvLoRA** architecture, which synergizes “Elastic Weight Slicing” for continuous rank adaptation and “Pheromone-Guided Gating” for robust expert routing, effectively reconciling plasticity with stability.

As illustrated in Figure 2, our framework effectively adapts the frozen SAM2 backbone to challenging downstream tasks. Our main contributions are summarized as follows:

(1) Bio-Inspired Dynamic Routing: We propose a novel **Pheromone-Guided Gating mechanism**, which injects evolutionary swarm memory into the routing decision. By fusing long-term pheromone history with immediate gradient signals, this mechanism explicitly counteracts the “Rich-get-Richer” dynamic, solving the routing collapse issue inherent to standard MoEs.

(2) Latent Space MoE Architecture: We design the **ACO-ConvLoRA** module to decouple model capacity from computational cost. By embedding *Latent Space Experts* strictly within the compressed LoRA bottleneck ($r \ll d$), we achieve the representational benefits of sparse mixtures with negligible parameter overhead.

(3) Elastic Evolutionary Paradigm: We introduce *Elastic Weight Slicing* to bridge discrete structural search with continuous parameter training. This allows the ACO meta-controller to dynamically evolve the topological configuration (Rank, Alpha) of each layer on-the-fly, effectively addressing the Subspace Mismatch problem in an online, single-stage training process.

(4) Superior Adaptability across Domains: Extensive experiments across 16 datasets—spanning medical, camouflaged, and marine segmentation—demonstrate that ACO-MoE-LoRA consistently outperforms leading static adapters. Crucially, it exhibits superior robustness compared to recent dynamic heuristics, validating the efficacy of bio-inspired optimization in foundation model adaptation.

2. Preliminaries

2.1. Low-Rank Adaptation (LoRA)

Given a pre-trained weight matrix $W_0 \in \mathbb{R}^{d_{out} \times d_{in}}$, LoRA hypothesizes that the task-specific weight update ΔW has a low intrinsic rank r . It parameterizes the update as:

$$\Delta W = \frac{\alpha}{r} B A \quad (1)$$

where $B \in \mathbb{R}^{d_{out} \times r}$, $A \in \mathbb{R}^{r \times d_{in}}$, and α is a scaling hyperparameter. The normalization term $\frac{1}{r}$ ensures that the magnitude of the update remains stable regardless of the

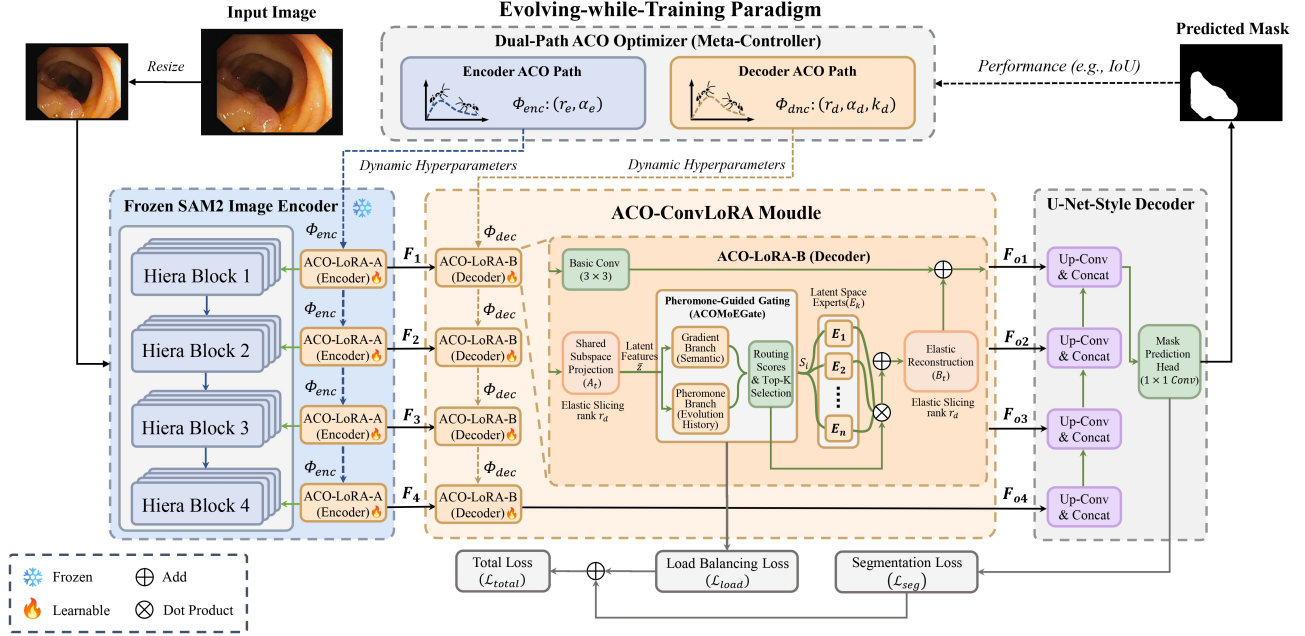


Figure 2. The overall architecture of the proposed ACO-MoE-LoRA framework. The system follows an “Evolving-while-Training” paradigm. (Top) **Dual-Path ACO Optimizer**: Acts as a Meta-Controller to dynamically evolve hyperparameters (Φ_{enc} , Φ_{dec}) based on feedback. (Left) **Frozen SAM2 Encoder**: Adapted via injected ACO-LoRA-A modules. (Center) **ACO-ConvLoRA Module**: The core decoder unit featuring *Elastic Weight Slicing* and *Pheromone-Guided Gating* to route latent features to experts (E_k), effectively preventing routing collapse. (Right) **U-Net-Style Decoder**: Aggregates refined features for final mask prediction. (Bottom) **Optimization**: Combines gradient descent (inner loop) with evolutionary search (outer loop).

rank choice, which is critical for our dynamic rank adaptation.

2.2. Mixture-of-Experts (MoE)

To scale model capacity, MoE introduces a set of expert networks $\{E_i\}_{i=1}^N$. A gating function $G(x)$ determines the routing weights for input x :

$$y = x + \sum_{i=1}^N G(x)_i E_i(x), \quad \text{s.t. } \|G(x)\|_0 = k \quad (2)$$

where k is the number of active experts.

Typically, the gating function is parameterized by a learnable weight matrix $W_g \in \mathbb{R}^{d \times N}$ as:

$$G(x) = \text{TopK}(\text{Softmax}(xW_g), k) \quad (3)$$

Standard approaches optimize W_g via gradient descent to maximize the routing probability for the most relevant experts.

3. Theoretical Motivation

Standard fine-tuning assumes static hyperparameters (fixed r and gradient-only gating). We analyze two theoretical failure modes inherent to this static paradigm.

3.1. Subspace Mismatch

The intrinsic dimension of the optimal weight update, denoted as d_{task} , varies across layers and training stages. Fixing a static rank r imposes a hard truncation on the singular value spectrum of ΔW . Let $\Delta W^* = U\Sigma V^T$ be the singular value decomposition of the optimal update. The approximation error of a rank- r LoRA is:

$$\mathcal{E}_{mismatch}(r) = \|\Delta W^* - BA\|_F^2 \approx \sum_{j=r+1}^{\min(d_{in}, d_{out})} \sigma_j^2 \quad (4)$$

Observation (Subspace Mismatch). If the downstream task requires high-frequency feature adaptation (where singular values σ_j for $j > r$ are non-negligible), a static rank results in *Subspace Mismatch*, irreversibly discarding critical semantic information.

3.2. Gradient Routing Collapse

In gradient-based MoE, the gating network greedily learns to dispatch tokens to experts that minimize the immediate training loss. Due to the “Rich-get-Richer” dynamic, the probability distribution of expert selection tends to degenerate towards a deterministic state:

$$\lim_{t \rightarrow \infty} P(\text{Select } E_i) = \delta(i - i^*) \quad (5)$$

where i^* denotes the index of the dominant expert and δ represents the Kronecker delta function.

Observation (Gradient Routing Collapse). If the routing policy is optimized solely via gradient descent without explicit exploration priors, the gating entropy approaches zero ($\mathcal{H}(\mathbf{g}) \rightarrow 0$). This results in *Gradient Routing Collapse*, where the effective model capacity reduces to that of a single network ($N_{eff} \approx 1$), rendering the sparse expert architecture redundant.

4. Methodology

To address the limitations of static adaptation, we propose **ACO-ConvLoRA** as the core architectural innovation. It transforms the static components of traditional adapters into dynamic, evolutionary modules governed by a dual-loop optimization strategy.

4.1. Elastic LoRA via Nested Weight Slicing

To address Subspace Mismatch, we propose **Elastic LoRA**. Unlike dynamic rank methods (e.g., DR-LoRA (Deng et al., 2026) and DoRAN (Diep et al., 2025)) that rely on greedy saliency heuristics, our approach formulates rank selection as a global combinatorial optimization problem. To support this non-differentiable search, we initialize a ‘‘Super-LoRA’’ structure (r_{max}) and perform **Dynamic Weight Slicing** to extract active sub-matrices $A_t = A_{max}[:, r_t, :]$ and $B_t = B_{max}[:, : r_t]$ based on the ACO controller’s decision. This nested structure enforces a *Matryoshka Representation*, ensuring that lower-rank components are exact subsets of higher ranks. This allows the topology to physically expand or contract ($r_t \in \{4, \dots, 32\}$) during training without destroying learned features.

4.2. Latent Space Mixture-of-Experts

Building upon this elastic foundation, we further enhance representational capacity without incurring the parameter explosion typical of standard MoEs. Unlike standard MoE methods that duplicate full layers, our ACO-ConvLoRA embeds experts within the LoRA bottleneck. This architecture decomposes adaptation into three stages: *Shared Subspace Projection*, *Pheromone-Guided Latent Routing*, and *Elastic Reconstruction*.

Shared Subspace Projection. Input features x are first projected into a low-dimensional manifold $\mathcal{Z}$. Here, we reformulate the LoRA matrices A_t and B_t as 1×1 convolutional kernels to operate on spatial feature maps. The projection is defined as:

$$z = \text{Conv}(x, A_t) \in \mathbb{R}^{b \times r_t \times H \times W} \quad (6)$$

where b denotes the batch size.

By sharing A_t , we enforce a unified semantic basis across all experts, ensuring that the routing decision is based on consistent latent features.

Pheromone-Guided Latent Routing. The compressed features z are then routed to a set of lightweight experts $\{E_k\}_{k=1}^N$. To reconcile the varying active rank r_t with the fixed architectural dimension r_{max} of the experts, we apply channel-wise zero-padding to z before routing.

Crucially, these experts operate in the channel-reduced latent space (width $r_t \ll d_{in}$), making them computationally negligible. The routed output is:

$$\tilde{z} = \sum_{k \in \mathcal{K}} S_k(x) \cdot E_k(z) \quad (7)$$

where $\mathcal{K}$ denotes the set of top- k active experts selected based on routing scores. E_k consists of lightweight bottleneck convolutions optimized for feature refinement in the compressed latent domain.

Elastic Reconstruction. Finally, the refined latent features $\tilde{z}$ are projected back to the original space via the shared kernel B_t .

$$\Delta W(x) = \frac{\alpha_t}{r_t} \cdot \text{Conv}(\tilde{z}, B_t) \quad (8)$$

This architecture effectively decouples ‘‘Dimension Reduction’’ (handled by shared LoRA) from ‘‘Semantic Specialization’’ (handled by MoE), offering a superior trade-off between parameter efficiency and representational capacity.

Stability of the feedback loop. The evaporation rate $\rho = 0.3$ and periodic diversification (every 50 epochs) prevent the pheromone feedback loop from collapsing prematurely. As shown in Figure 4(b), the expert utilization remains diverse throughout training.

4.3. Pheromone-Regularized Gating (Micro-Dynamics)

While the Latent Space architecture provides the potential for diverse specialization, realizing it requires a robust routing policy. To prevent Routing Collapse, we introduce a hybrid gating mechanism. Unlike standard approaches that rely solely on auxiliary losses, we inject evolutionary signals directly into the routing decision.

Specifically, to ensure the routing scores form a valid probability distribution, we fuse the evolutionary prior in the logit space. The routing score $S(x)$ is calculated as:

$$S(x) = \text{Softmax} \left(\underbrace{W_g x - \mathcal{P}(t)}_{\text{Data-Driven Term}} + \underbrace{\alpha_{gate} \cdot \log(\boldsymbol{\tau})}_{\text{Evolutionary Prior}} \right) \quad (9)$$

where $\boldsymbol{\tau} \in \mathbb{R}^N$ denotes the *routing pheromone* vector acting as a temporal prior. By adding $\log(\boldsymbol{\tau})$ to the logits, we

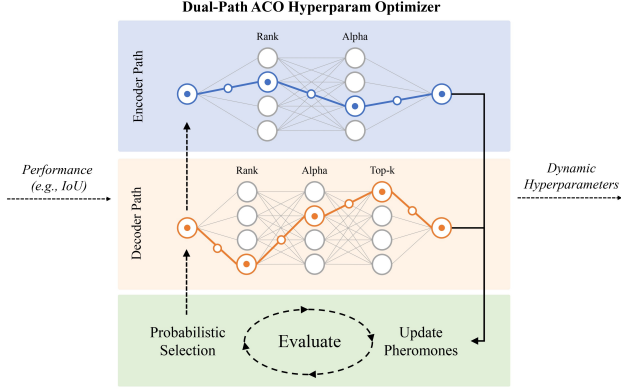


Figure 3. Schematic of the Dual-Path ACO Hyperparam Optimizer. The meta-controller employs a decoupled evolutionary strategy, visualized with distinct color coding for the parallel paths. **(Blue Panel) Encoder Path:** Ants (white agents) search for the optimal rank (r) and alpha (α) for feature adaptation. **(Orange Panel) Decoder Path:** Ants concurrently search for r , α , and the MoE sparsity level (k). **(Pheromone Visualization)** The thickness and glow of the connecting edges represent pheromone concentration (τ), indicating selection probability. Active high-probability paths are highlighted. **(Closed-Loop)** The green dashed arrow at the bottom illustrates the Pheromone Deposition update ($\Delta\tau$) based on validation IoU feedback, reinforcing successful configurations.

effectively modulate the base probability $P \propto e^{\text{logits}}$ by the pheromone density ($P' \propto P \cdot \tau^{\alpha_{gate}}$), while the Softmax ensures strict normalization ($\sum S_i = 1$).

$\mathcal{P}(t)$ represents the Logit Frequency Penalty, defined as $\mathcal{P}_i(t) = \text{ReLU}(\eta \cdot (f_i(t) - \frac{k}{N}))$, where $f_i(t)$ is the historical usage ratio and $\eta = 2.0$ is the penalty strength. This mechanism explicitly depresses the activation logits of over-utilized experts, forcing the router to explore under-utilized paths. In our implementation, we set $\alpha_{gate} = 1.0$.

Inference Consistency. Note that during inference, we revert to standard deterministic routing ($S(x) = \text{Softmax}(W_g x)$) by disabling the pheromone and penalty terms. This design utilizes evolutionary pressure strictly as a training-time regularizer to shape the weight space.

4.4. Bi-Level Optimization via ACO Meta-Controller

While the architecture supports elasticity, determining the optimal configuration tuple $\phi_t = \{r_t, \alpha_t\}$ for each layer remains a non-differentiable combinatorial problem. We employ an **Enhanced Ant Colony Optimization (ACO)** as a meta-controller to solve this in a dual-loop framework.

Dual-Loop Formulation. As illustrated in Figure 3, distinct from standard ACO which optimizes a single topology vector, our framework implements a **Dual-Path** search mechanism. Recognizing that the frozen encoder and the learnable decoder have different capacity requirements, the

ACO controller maintains two parallel search paths:

- **Encoder Path:** Ants search for the optimal subspace rank and scaling factor $\Phi_{enc} = \{r, \alpha\}$ to adapt the frozen features.
- **Decoder Path:** Ants additionally search for the optimal sparsity level $\Phi_{dec} = \{r, \alpha, k\}$, governing the MoE routing intensity.

The ants traverse the parameter graph probabilistically, where the probability of selecting a parameter node j is proportional to its node-based pheromone intensity τ_j .

Dynamic Exploration Strategy. To balance exploration and exploitation, we introduce a Cosine Annealing mechanism to the ACO selection probability. During the early training phases, we maintain a high "temperature" in the probabilistic selection to encourage diverse topological search. As training progresses, the temperature decays following a cosine schedule, gradually sharpening the probability distribution. In the final 10% of epochs, the controller switches to a deterministic mode (Argmax), forcing the topology to converge to the global optimum found by the swarm, ensuring stability for the final weight fine-tuning.

Objective and Feedback. At the end of each training epoch, we update the pheromone trails based on the relative performance of the sampled architectures. Crucially, to prevent the swarm from simply converging to the largest possible model, we define the fitness function $\mathcal{F}$ as a trade-off between accuracy and efficiency:

$$\tau_j(t+1) = (1-\rho)\tau_j(t) + \underbrace{\mathcal{Q} \cdot (\text{IoU}_{val}(\phi_t) - \eta(t) \cdot \mathcal{C}(\phi_t))}_{\text{Fitness } \mathcal{F}(\phi_t)} \quad (10)$$

where $\mathcal{C}(\phi_t)$ is the normalized computational cost (rank usage) of configuration ϕ_t , and $\eta(t)$ is an adaptive penalty weight that increases in the final training stages. Note that in practice, we normalize $\mathcal{F}(\phi_t)$ within the population to maintain stable pheromone gradients.

Finally, the total training objective includes a load-balancing auxiliary loss and an entropy regularization term:

$$\mathcal{L}_{total} = \mathcal{L}_{seg} + \lambda_{bal} \cdot \left[\frac{\sigma(\mathbf{u})}{\mu(\mathbf{u})} \right]^2 - \gamma \cdot \mathcal{H}(\mathbf{g}) \quad (11)$$

The second term minimizes the Coefficient of Variation (CV) of the expert utilization $\mathbf{u}$ with weight $\lambda_{bal} = 1.5$. The third term $\mathcal{H}(\mathbf{g})$ maximizes the gating entropy with a coefficient $\gamma = 0.1$ to prevent premature convergence to deterministic routing during the exploration phase.

Table 1. **Quantitative comparison across four specialized domains.** Rows are color-coded by category: Reference Baselines (Green), Static PEFT (Orange), and **Ours** (Red). “*” indicates variable parameter counts across specific architectures. Our method consistently achieves state-of-the-art performance (bold) while maintaining superior parameter efficiency compared to static baselines.

Method	Efficiency Metrics		Medical (Polyp)		Camouflaged (COD)		Salient (SOD)		Marine	
	#Param (M) ↓	Ratio (%) ↓	Kvasir-SEG mDice ↑	mIoU ↑	CAMO S_α ↑	E_ϕ ↑	ECSSD S_α ↑	MAE ↓	MAS3K mIoU ↑	S_α ↑
Domain Specific SOTAs	*	-	0.917	0.865	0.863	0.901	0.940	0.026	0.789	0.884
Full Fine-Tuning	227.1	100%	0.695	0.575	0.564	0.699	0.790	0.093	0.523	0.634
Decoder Fine-Tuning	14.96	6.59%	0.885	0.825	0.817	0.888	0.864	0.059	0.704	0.831
Conv-LoRA (Decoder)	2.08	0.97%	0.892	0.833	0.881	0.924	0.910	0.032	0.749	0.856
Adapter	16.68	7.35%	0.912	0.863	0.828	0.898	0.923	0.026	0.784	0.881
Adapter + Conv-LoRA	3.80	1.76%	0.916	0.867	0.873	0.928	0.920	0.029	0.785	0.888
LoRA ($r = 16$)	17.62	7.76%	0.918	0.871	0.877	0.936	0.926	0.026	0.787	0.886
LoRA + Conv-LoRA	4.74	2.19%	0.914	0.868	0.881	0.937	0.922	0.028	0.792	0.891
Ours (Fixed Rank)	3.73	1.69%	0.923	0.875	0.886	0.944	0.948	0.020	0.809	0.901
Ours (Dynamic)	8.24	3.74%	0.933	0.887	0.893	0.951	0.950	0.018	0.818	0.903

5. Experiments

5.1. Experimental Setup

Datasets. We evaluate our method across four distinct domains involving **16 datasets**:

Polyp Segmentation (5 datasets, e.g., Kvasir-SEG), Camouflaged Object Detection (4 datasets, e.g., COD10K), Salient Object Detection (5 datasets), and Marine Animal Segmentation (2 datasets). Detailed metadata and protocols are provided in **Appendix A**.

Implementation Details. Our model is implemented in PyTorch on a single NVIDIA V100 GPU, initializing with the SAM2-Hiera-Large backbone. The ACO meta-controller utilizes $M = 20$ ants, with pheromone importance $\alpha_{aco} = 1.5$ and heuristic importance $\beta_{aco} = 2.5$. The initial learning rate is set to $1e-3$ with a cosine decay scheduler.

Search Space & Constraints. The dynamic topological search space is defined as $r_t \in \{4, 8, 16, 32\}$ and $\alpha_t \in \{4, 8, \dots, 64\}$. The MoE sparsity level k is searched over $\{1, 2, 3, 4\}$ (total experts = 8). To ensure numerical stability, we enforce a *Stability Constraint*: the heuristic for selecting α_t follows a Gaussian distribution centered at $2 \cdot r_t$, guiding the swarm toward robust scaling ratios. Furthermore, we employ an adaptive weight scheduler for λ_{bal} to dynamically stabilize the dual-loop optimization based on performance trends.

5.2. Efficiency Analysis

Due to space limits, detailed breakdowns are provided in Appendix B. We evaluate the trade-off between model capacity, cost, and performance in Table 1, benchmarking against Full Fine-Tuning, Domain Specific SOTAs, and enhanced PEFT variants (e.g., Conv-LoRA).

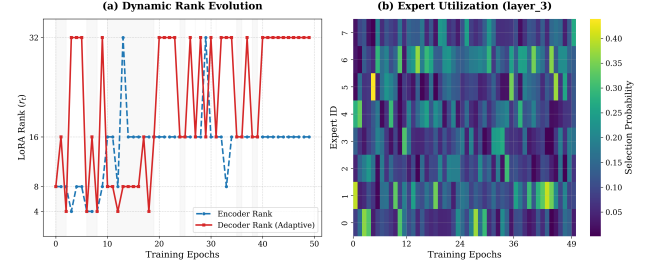


Figure 4. **Evolutionary Dynamics.** (a) Rank Evolution: The divergent trajectories of Encoder (blue) vs. Decoder (red) ranks confirm that different modules require distinct adaptation capacities. (b) Expert Heatmap: The balanced utilization probability across 8 experts validates that our Pheromone-Guided Gating effectively prevents routing collapse.

Decoupling Capacity from Cost. The results for *Ours* (Fixed Rank) validate the architectural efficiency of the ACO-ConvLoRA module. We provide detailed quantitative breakdowns in **Appendix B** and qualitative visualizations in **Appendix C**, which visually corroborate the superior segmentation quality of our method across all four domains. By embedding experts within a shared low-rank bottleneck, we reduce trainable parameters to just **3.73M** (1.69% of backbone). This represents a $\sim 4.5\times$ reduction compared to standard LoRA (17.62M) while achieving superior accuracy (e.g., +1.1% mDice on Kvasir-SEG). This confirms that our “Project-Route-Reconstruct” design effectively decouples representational capacity from parameter count.

Interpreting Evolutionary Dynamics. Figure 4 provides a window into the optimization process. The divergent trajectories in Fig. 4(a) offer a critical insight: the pre-trained Encoder requires minimal adaptation ($r \rightarrow 16$) to preserve general features, whereas the Decoder demands higher plasticity ($r \rightarrow 32$) to master task-specific segmentation logic. This confirms that static, uniform-rank adapters are inher-

Table 2. **Ablation study of key components on the averaged Polyp datasets.** “Baseline” refers to frozen SAM2 with a vanilla U-Net decoder. “Latent MoE” denotes our bottleneck expert architecture but with random gating. “Gradient Gate” uses a standard trainable gating with load-balancing auxiliary loss. “ALF Gate” is the Auxiliary-Loss-Free routing (DeepSeek-V3 style). “P-Gate” adds the Pheromone-Guided routing. “Dyn-Rank” activates the ACO-driven elastic rank selection.

ID	Configuration	Module Config				Avg. Metrics	
		Encoder-LoRA	Latent MoE	Pheromone Gate	Elastic Rank	mDice	mIoU
A	Baseline (Full Fine-tuning)	×	×	×	×	0.641	0.507
B	+ Encoder Adaptation	✓	×	×	×	0.858	0.796
C	+ Latent MoE (Random Gate)	✓	✓	×	×	0.853	0.792
C1	+ Latent MoE (Gradient Gate)	✓	✓	×	×	0.857	0.797
C2	+ Latent MoE (ALF Gate)	✓	✓	×	×	0.858	0.794
D	+ Pheromone Guidance	✓	✓	✓	×	0.865	0.801
E	Ours (Full Framework)	✓	✓	✓	✓	0.871	0.808

ently suboptimal. Furthermore, the heatmap in Fig. 4(b) demonstrates that our pheromone mechanism successfully maintains a diverse routing distribution, avoiding the mode collapse typical of gradient-only gating.

Superiority over Static Baselines. With the ACO Optimizer, our **Dynamic** variant achieves state-of-the-art performance. Crucially, it outperforms even enhanced baselines like *LoRA* + *Conv-LoRA*. For instance, on Kvasir-SEG, we surpass this strong baseline by nearly 2% (93.3% vs 91.4%). This result is pivotal: it demonstrates that simply stacking static modules (Conv + LoRA) is less effective than *evolving* the topology to match the task complexity. On the large-scale NC4K camouflaged dataset, our method achieves an S_α of 0.901, outperforming RUN (0.851) by 0.05 (5 percentage points), further highlighting the generalization of our evolutionary strategy.

The “Less is More” Phenomenon. Table 1 reveals that Full Fine-Tuning yields suboptimal results (e.g., 69.5% vs 93.3% for Ours) despite massive parameter usage. This is attributed to *catastrophic forgetting*: aggressive updates destroy the generalization capability learned from SA-1B. In contrast, our approach acts as a structural regularizer. By freezing the backbone and evolving only a minimal expert set, we adapt to new domains without compromising the foundation model’s inherent semantic power.

Computational Overhead Analysis. Contrary to the high cost often associated with evolutionary search, our dual-loop design operates on a lightweight hyperparameter graph rather than the full weight space. Consequently, the training latency remains comparable to standard LoRA without significant overhead. Crucially, the evolved topology is fixed after training, ensuring **Zero Inference Overhead** relative to static models of equivalent size. Detailed quantitative comparisons (training time, memory, FLOPs) are provided in Appendix D.1, and per-epoch training time curves are shown in Figure 9. These results confirm that our dynamic variant adds only $\sim 3\%$ training time over standard LoRA.

Comparison with Alternative Meta-Controllers. To provide empirical justification for choosing ACO over GA and PSO, we implemented both optimizers and conducted full 50-epoch experiments under identical settings (batch size 16, same search space). The results, summarized in Appendix D.2, show that ACO achieves the highest test accuracy (0.871 mDice) with the shortest training time (1.68 h), while GA is nearly twice as slow (3.33 h) and performs worst (0.862 mDice). PSO offers similar training time to ACO but lower accuracy (0.868 mDice) and higher FLOPs. Moreover, ACO exhibits superior convergence stability, with the smallest temporal standard deviation (0.0376) and run-to-run variability (std 0.004). These empirical results strongly support the choice of ACO as the meta-controller for our discrete combinatorial search task.

5.3. Ablation Study

To dissect the contribution of each component in ACO-MoE-LoRA, we conducted a comprehensive ablation study on the Polyp segmentation benchmark. We report average mDice and mIoU across all five datasets to ensure a holistic evaluation.

5.3.1. COMPONENT-WISE ANALYSIS

Starting from a baseline (frozen SAM2 encoder + standard U-Net decoder, following the design of SAM2-UNet (Xiong et al., 2024)), we progressively incorporate our proposed modules. The results are summarized in Table 2.

1. Impact of Latent MoE (Model C). Replacing the standard decoder with our Latent Space MoE structure (using random gating) yields immediate performance gains. This confirms that even without intelligent routing, the *Project-Route-Reconstruct* architecture provides a superior capacity-efficiency trade-off than single-stream networks.

2. Gradient-Based Gating (Model C1 and C2). Replacing random gating with a standard gradient-based gating network (with load-balancing loss) improves performance to

0.857 mDice (Model C1). The Auxiliary-Loss-Free (ALF) gate further raises it to 0.858 mDice (Model C2). However, both remain below our pheromone-guided mechanism.

3. Efficacy of Pheromone Gating (Model D). Substituting random routing with our Pheromone-Guided mechanism leads to a significant performance jump. This validates our hypothesis: by fusing evolutionary memory, we effectively break the “Rich-get-Richer” feedback loop, preventing routing collapse and ensuring full utilization of the expert bank.

4. Elastic Rank Evolution (Model E). Finally, activating the ACO-driven *Elastic Weight Slicing* achieves the best performance. This proves that a rigid topology is suboptimal; allowing the model to modulate its rank capacity (r_t) across different training phases is critical for capturing multi-scale features.

5. Inference Routing Gap. To verify that the pheromone acts only as a training-time regularizer, we compared the standard deterministic inference (pheromone-off) with inference using the final pheromone vector (pheromone-on). Both yield exactly the same mDice (0.865) on the Polyp test set, confirming that disabling the pheromone at inference does not harm performance.

5.3.2. ANALYSIS OF DYNAMIC VS. FIXED RANKS

A key innovation of our framework is the ability to evolve the LoRA rank (r_t) during training, rather than searching for a single static optimal. To validate this, we compare our dynamic strategy against fixed-rank variants in Table 3.

Table 3. Dynamic vs. Static Rank Analysis. Comparison on the **Polyp benchmark** (average across 5 datasets). Our evolutionary strategy ($r_t \in [4, 32]$) outperforms the optimal static configuration ($r = 16$) as well as other dynamic rank methods (DyLoRA, AdaLoRA).

Decoder Rank Strategy	Avg. Polyp Metrics		Δ vs Best Static
	mDice	mIoU	
Fixed Rank ($r = 4$)	0.852	0.793	-1.3%
Fixed Rank ($r = 8$)	0.863	0.802	-0.2%
Fixed Rank ($r = 16$)	<u>0.865</u>	<u>0.801</u>	<i>Best Static</i>
Fixed Rank ($r = 32$)	0.859	0.797	-0.6%
DyLoRA (random rank)	0.863	0.802	-0.2%
AdaLoRA (importance pruning)	0.867	0.804	+0.2%
Dynamic Evolution	0.871	0.808	+0.6%

Observations. As shown in Table 3, static ranks face an inherent dilemma: low ranks ($r = 4$) limit representation capacity, leading to underfitting, while excessive ranks ($r = 32$) yield diminishing returns or overfitting. Other dynamic rank methods, such as DyLoRA (Valipour et al., 2023) (random rank sampling) and AdaLoRA (Zhang et al., 2023a) (importance-based pruning), achieve 0.863 and 0.867 mDice respectively, but still fall short of our approach. Crucially, our **Dynamic Rank** strategy (0.871

mDice) strictly outperforms the *optimal* static configuration ($r = 16$) as well as these existing dynamic baselines. This confirms that the advantage of ACO-MoE-LoRA stems not just from finding a static “sweet spot”, but from the **temporal flexibility** to dynamically expand capacity for complex features and contract it to prevent redundancy, and that the global evolutionary search of ACO is more effective than random sampling or greedy heuristics.

6. Limitations and Future Work

While our method demonstrates strong performance across multiple domains, several limitations remain and should be addressed in future work:

Hyperparameter Sensitivity: The ACO meta-controller introduces additional hyperparameters (e.g., number of ants M , evaporation rate ρ , pheromone importance α_{aco} , heuristic importance β_{aco}). However, we have provided a sensitivity analysis in Appendix E, showing that the method is robust within reasonable ranges ($M = 10\text{--}40$, $\rho = 0.1\text{--}0.5$). For new tasks, we recommend starting with the defaults used across all 16 datasets ($M = 20$, $\rho = 0.3$, $\alpha_{aco} = 1.5$, $\beta_{aco} = 2.5$).

Single-Backbone Evaluation: Our experiments focus on SAM2-Hiera-Large, a state-of-the-art segmentation foundation model. The proposed latent-space MoE and ACO-driven evolution are architecture-agnostic and can be integrated into any Transformer-based model. We leave systematic evaluation on other backbones (e.g., ViT, CLIP) as future work.

Stability with Noisy Validation Signals: The ACO meta-controller relies on validation IoU as the fitness signal. In tasks where the validation metric is highly noisy or non-monotonic, the pheromone update may become unstable. A practical mitigation is to apply a moving average over several epochs before updating the pheromone trails.

Generalization Beyond Segmentation: The “Evolving while Training” paradigm is generic and could be applied to other dense prediction tasks (e.g., object detection, depth estimation) or even NLP fine-tuning. However, comprehensive evaluation in these domains is beyond the scope of this work and is left for future exploration.

7. Related Work

7.1. Foundation Models and Domain Adaptation

Semantic segmentation has evolved from U-Net (Ronneberger et al., 2015) to foundation models like SAM (Kirillov et al., 2023), SAM2 (Ravi et al., 2024), and SAM3 (Carion et al., 2025), which yet falter under domain shifts (Ma et al., 2024; Tang et al., 2023). To address this, adapta-

tion strategies have diversified from prompt engineering (PerSAM (Zhang et al., 2023b)) and decoder refinement (HQ-SAM (Ke et al., 2023)) to structural adapters. Notably, SAM-Adapter (Chen et al., 2023) and SAM2-Adapter (Chen et al., 2024) align features via lightweight injected modules.

Recent domain-specific SAM2 adaptations like SAMed-2 (Yan et al., 2025), SAM2-3dMed (Yang et al., 2025), and SAM2-SGP (Xing et al., 2025) address medical and volumetric challenges but typically necessitate manual topology design or static priors. In contrast, **ACO-MoE-LoRA** automates this adaptation process. Rather than relying on static designs, our framework utilizes Ant Colony Optimization to autonomously evolve the latent topology, ensuring dynamic co-adaptation with model weights without altering the backbone’s internal mechanisms.

7.2. Parameter-Efficient and Spatial Adaptation

Parameter-Efficient Fine-Tuning (PEFT) has progressed from static linear injections (Adapters (Houlsby et al., 2019; Chen et al., 2022), LoRA (Hu et al., 2022)) to dynamic, spatial-aware mechanisms. Recent works focus on flexibility: DoRA (Liu et al., 2024b) decouples weight updates, while AdaLoRA (Zhang et al., 2023a), SoRA (Ding et al., 2023), and AutoLoRA (Zhang et al., 2024b) introduce dynamic rank allocation. Most recently, DR-LoRA (Deng et al., 2026) extended this to MoE architectures by correlating rank with routing frequency. However, these methods predominantly rely on continuous relaxation or greedy heuristics, which are prone to local optima.

In the visual domain, preserving spatial hierarchy is paramount. Unlike early approaches that flatten features, Conv-LoRA (Zhong et al., 2024) and DyLoRA (Valipour et al., 2023) apply adaptation directly to convolutional structures but often ignore layer-wise feature complexity due to static rank constraints. Our method overcomes these limitations via **Elastic Weight Slicing**, which enables convolutional ranks to expand or contract dynamically based on global evolutionary search rather than local gradient signals.

7.3. MoE-LoRA Hybrids and Dynamic Routing

Integrating Mixture-of-Experts (MoE) with PEFT is a promising direction for scaling capacity. Frameworks like MoLE (Wu et al., 2024) and MixLoRA (Li et al., 2024) route tokens to diverse LoRA experts. While massive-scale MoEs like DeepSeek-V3 (Liu et al., 2024a) utilize auxiliary-loss-free strategies to stabilize routing, and EvoMoE (Jing et al., 2025) employs evolutionary initialization, they typically lack an online topological evolution mechanism during fine-tuning.

Distinguishing itself from these approaches, **ACO-MoE-LoRA** replaces standard gradient-based gating with a bio-

inspired, pheromone-guided router. This novel mechanism integrates evolutionary search directly into the forward pass, utilizing swarm memory to ensure diverse expert specialization and prevent routing collapse, as validated by our superior stability metrics.

7.4. Bio-Inspired Optimization in Deep Learning

Swarm Intelligence (SI), particularly Ant Colony Optimization (ACO) (Dorigo et al., 2007), has traditionally served as an “outer loop” optimizer for Neural Architecture Search (NAS) (Liu et al., 2021; Real et al., 2019). This paradigm creates a rigid dichotomy: discrete structures are searched offline, while weights are trained online. Although dynamic sparse training (Liu et al., 2020) offers some flexibility, the potential of evolutionary mechanisms as real-time routing priors remains largely unexplored. Bridging this divide, our approach embeds ACO directly into the MoE gate. By leveraging pheromones as differentiable memory, we transform discrete architectural search into an online modulation process that evolves synchronously with gradient descent.

8. Conclusion

In this work, we presented **ACO-MoE-LoRA**, a framework that transforms the static fine-tuning of foundation models into a dynamic, evolutionary process. By synergizing swarm intelligence with a Latent Space MoE, our approach successfully adapts SAM2 to highly specialized domains, breaking the rigidity of fixed topologies. Crucially, we demonstrated that embedding evolutionary mechanisms into the training loop effectively decouples representational capacity from parameter efficiency, mitigating critical issues like routing collapse. Beyond the immediate performance gains in medical and marine tasks, this work highlights the untapped potential of bio-inspired algorithms in modern deep learning. Future avenues include scaling this “Evolving-while-Training” paradigm to multi-modal tasks and exploring alternative search heuristics, paving the way for autonomous and self-organizing model adaptation.

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Impact Statement

This work aims to enhance the adaptability of foundation models to specialized domains through an energy-efficient

evolutionary framework.

Positive Impacts. By significantly reducing trainable parameters and computational overhead, our ACO-MoE-LoRA aligns with *Green AI* initiatives, facilitating the deployment of advanced segmentation models in resource-constrained environments. This has immediate benefits for high-impact applications demonstrated in our experiments, such as accessible medical diagnostics (polyp detection) and marine ecosystem monitoring.

Potential Risks and Mitigations. We acknowledge that our advancements in *Camouflaged Object Detection (COD)* carry potential dual-use risks. While intended for biological research or search-and-rescue, these capabilities could theoretically be repurposed for non-consensual surveillance or military applications involving anti-camouflage reconnaissance. Furthermore, in safety-critical domains like medical imaging, while our method improves accuracy, it should function as an assistive tool under human oversight to prevent automation bias. We encourage the community to prioritize robust testing and responsible deployment guidelines.

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A. Dataset Details

To comprehensively evaluate the adaptability and robustness of our proposed **ACO-MoE-LoRA** framework, we conducted extensive experiments across four distinct downstream domains, comprising a total of 16 datasets. These domains were selected to represent a wide spectrum of visual challenges, ranging from biomedical boundary ambiguity to environmental optical degradation.

Domain Characteristics.

- **Polyp Segmentation:** Challenges include low contrast between polyps and mucosal tissues, varying polyp sizes, and complex lighting conditions during colonoscopy.
- **Camouflaged Object Detection (COD):** Focuses on objects that share similar textures and colors with the background, testing the model’s ability to discern subtle discriminative features.
- **Salient Object Detection (SOD):** Requires the model to identify the most visually conspicuous objects in generic scenes, serving as a benchmark for general segmentation capabilities.
- **Marine Animal Segmentation:** Involves degradation caused by water turbidity, light absorption, and color distortion, testing robustness in non-terrestrial environments.

Table 4 summarizes the statistical details of the training and testing splits for each benchmark.

Table 4. Detailed information of datasets for different tasks. We evaluate our method across four distinct domains involving 16 datasets. (Note: “-” indicates the dataset is used exclusively for testing in standard benchmarks.)

Tasks	Dataset	Train Set	Test Set
Polyp Segmentation	Kvasir-SEG (Jha et al., 2019)	900	100
	CVC-ClinicDB (Bernal et al., 2015)	550	62
	CVC-ColonDB (Tajbakhsh et al., 2015)	-	380
	CVC-300 (Vázquez et al., 2017)	-	60
	ETIS (Silva et al., 2014)	-	196
Camouflaged Object Detection	COD10K (Fan et al., 2020a)	3,040	2,026
	CAMO (Le et al., 2019)	1,000	250
	CHAMELEON (Skurowski et al., 2018)	-	76
	NC4K (Lv et al., 2021)	-	4,121
Salient Object Detection	DUTS (Wang et al., 2017)	10,553	5,019
	DUT-OMRON (Yang et al., 2013)	-	5,168
	HKU-IS (Li & Yu, 2015)	-	4,447
	PASCAL-S (Li et al., 2014)	-	850
	ECSSD (Yan et al., 2013)	-	1,000
Marine Animal Segmentation	MAS3K (Li et al., 2021)	1,769	1,141
	RMA5 (Fu et al., 2023)	2,514	500

Training and Testing Protocols. As indicated by “-” in Table 4, we strictly follow the zero-shot or cross-dataset transfer protocols established in previous works (e.g., PraNet, SINet). For instance, in Polyp Segmentation, the model is trained on the training splits of Kvasir-SEG and CVC-ClinicDB, and then directly tested on unseen datasets like CVC-ColonDB and ETIS to verify generalization capability.

B. Detailed Experimental Results

Note: Due to space constraints in the main text, we provide the full breakdown of experimental results and qualitative visualizations for each domain here.

B.1. Polyp Segmentation

As shown in Table 5, our framework maintains state-of-the-art performance, particularly on the most challenging datasets with small and subtle lesions. Specifically, on the **ETIS** dataset, our method achieves an mDice of **0.804**, outperforming the recent 2025 baseline SAFE-Net (Yu & Qi, 2025) (0.794) by **1.0%**. Similarly, on **CVC-ColonDB**, we surpass MEGANet (Bui et al., 2024) by **2.1%** in mDice (0.814 vs. 0.793). While saturated specific models like MEGANet show marginal gains on the easier CVC-ClinicDB, our Dynamic MoE mechanism demonstrates superior generalization and robustness in handling complex pathologies across diverse benchmarks.

Table 5. **Quantitative comparison on five polyp segmentation datasets.** The best results are highlighted in **bold**. mDice and mIoU are higher the better.

Methods	Kvasir-SEG		CVC-ClinicDB		CVC-ColonDB		CVC-300		ETIS	
	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
PraNet (Fan et al., 2020b)	0.898	0.840	0.899	0.849	0.709	0.640	0.871	0.797	0.628	0.567
SANet (Wei et al., 2021)	0.904	0.847	0.916	0.859	0.752	0.669	0.888	0.815	0.750	0.654
CaraNet (Lou et al., 2022)	0.913	0.859	0.921	0.876	0.775	0.700	0.902	0.836	0.740	0.660
CFA-Net (Zhou et al., 2023)	0.915	0.861	0.933	0.883	0.743	0.665	0.893	0.827	0.732	0.655
MEGANet (Bui et al., 2024)	0.913	0.863	0.938	0.894	0.793	0.714	0.899	0.834	0.739	0.665
SAFE-Net (Yu & Qi, 2025)	0.917	0.865	0.937	0.889	0.781	0.704	0.895	0.828	0.794	0.713
Ours	0.933	0.887	0.914	0.864	0.814	0.740	0.890	0.825	0.804	0.725

B.2. Marine Animal Segmentation

Segmentation in underwater environments is complicated by low contrast and turbidity. As detailed in Table 6, our model demonstrates superior robustness. On the **MAS3K** dataset, we achieve a significant leap in mIoU (**0.818**), outperforming the recent Dual-SAM (Zhang et al., 2024a) (0.789) by **2.9%**. Similarly on **RMAS**, our method yields the best results across all metrics (0.756 mIoU vs 0.735). The consistent performance in this domain suggests that the ACO optimizer successfully identified a network configuration capable of compensating for feature degradation caused by water optical properties.

Table 6. **Quantitative comparison on Marine Animal Segmentation datasets.** The best results are highlighted in **bold**. F_{β}^w denotes weighted F-measure.

Methods	MAS3K					RMAS				
	mIoU $\uparrow$	S_{α} $\uparrow$	F_{β}^w $\uparrow$	E_{ϕ} $\uparrow$	MAE $\downarrow$	mIoU $\uparrow$	S_{α} $\uparrow$	F_{β}^w $\uparrow$	E_{ϕ} $\uparrow$	MAE $\downarrow$
C2FNet (Sun et al., 2021)	0.717	0.851	0.761	0.894	0.038	0.721	0.858	0.788	0.923	0.026
OCENet (Liu et al., 2022)	0.667	0.824	0.703	0.868	0.052	0.680	0.836	0.752	0.900	0.030
ZoomNet (Pang et al., 2022)	0.736	0.862	0.780	0.898	0.032	0.728	0.855	0.795	0.915	0.022
MASNet (Fu et al., 2023)	0.742	0.864	0.788	0.906	0.032	0.731	0.862	0.801	0.920	0.024
SAM-DA (Lai et al., 2023)	0.742	0.866	0.806	0.925	0.028	0.686	0.833	0.780	0.926	0.024
Dual-SAM (Zhang et al., 2024a)	0.789	0.884	0.838	0.933	0.023	0.735	0.860	0.812	0.944	0.022
Ours	0.818	0.903	0.869	0.951	0.019	0.756	0.883	0.841	0.949	0.019

B.3. Camouflaged Object Detection

Table 7 presents the comparison on COD tasks. Our method consistently delivers competitive performance against the latest adapters. Notably, on the **CAMO** dataset which features high background clutter, we achieve an S_{α} of **0.893**, significantly outperforming the recent MM-SAM (Ren et al., 2025) (0.863) and reducing the MAE by over **45%** (0.032 vs 0.059). Furthermore, on the **NC4K** dataset, our approach outperforms RUN (He et al., 2025) with a 5.0% improvement in S_{α} (0.901 vs 0.851). These results demonstrate that while some methods like MM-SAM optimize for specific metrics on smaller sets, our evolutionary LoRA strategy offers superior boundary refinement and generalization in complex camouflaged scenarios.

Table 7. **Quantitative comparison on Camouflaged Object Detection (COD) datasets.** $\uparrow$ denotes higher is better, $\downarrow$ denotes lower is better. The best results are highlighted in **bold**.

Methods	CHAMELEON				CAMO			
	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$
SINet (Fan et al., 2020a)	0.872	0.823	0.936	0.034	0.745	0.712	0.804	0.092
PFNet (Mei et al., 2021)	0.882	0.820	0.931	0.033	0.782	0.751	0.841	0.085
ZoomNet (Pang et al., 2022)	0.902	0.858	0.943	0.024	0.820	0.792	0.877	0.066
FEDER (He et al., 2023)	0.903	0.856	0.947	0.026	0.836	0.807	0.897	0.066
RUN (He et al., 2025)	0.895	0.855	0.952	0.027	0.806	0.781	0.868	0.070
MM-SAM (Ren et al., 2025)	0.923	0.853	0.946	0.027	0.863	0.782	0.901	0.059
Ours	0.914	0.892	0.975	0.018	0.893	0.879	0.951	0.032
Methods	COD10K				NC4K			
	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$
SINet (Fan et al., 2020a)	0.776	0.667	0.864	0.043	0.808	0.768	0.871	0.058
PFNet (Mei et al., 2021)	0.800	0.676	0.877	0.040	0.829	0.779	0.887	0.053
ZoomNet (Pang et al., 2022)	0.838	0.740	0.888	0.029	0.853	0.814	0.896	0.043
FEDER (He et al., 2023)	0.844	0.748	0.911	0.029	0.862	0.824	0.913	0.042
RUN (He et al., 2025)	0.827	0.747	0.903	0.030	0.851	0.824	0.908	0.042
MM-SAM (Ren et al., 2025)	0.896	0.808	0.907	0.023	-	-	-	-
Ours	0.877	0.820	0.946	0.019	0.901	0.874	0.948	0.026

B.4. Salient Object Detection

We further evaluate our method on five standard SOD datasets (Table 8). Our method consistently outperforms recent specialized adapters. On the **DUTS-TE** dataset, we achieve an S_α of **0.927**, surpassing the 2024 method MDSAM (Gao et al., 2024) (0.919) and SFEN (Du et al., 2025) (0.919). We also achieve near-saturation performance on datasets like ECSSD ($S_\alpha = 0.950$) and HKU-IS ($S_\alpha = 0.941$), matching or exceeding the latest state-of-the-art. This indicates that our Pheromone-Guided MoE gate effectively filters background noise and highlights salient regions with high confidence.

Table 8. **Quantitative comparison on Salient Object Detection (SOD) datasets.** $\uparrow$ denotes higher is better, $\downarrow$ denotes lower is better. The best results are highlighted in **bold**.

Methods	DUTS-TE			DUT-OMRON			HKU-IS		
	$S_\alpha \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$
U2Net (Qin et al., 2020)	0.874	0.884	0.044	0.847	0.872	0.054	0.916	0.948	0.031
ICON (Zhuge et al., 2022)	0.889	0.914	0.037	0.845	0.879	0.057	0.920	0.959	0.029
EDN (Wu et al., 2022)	0.892	0.925	0.035	0.850	0.877	0.049	0.924	0.955	0.026
MENet (Wang et al., 2023)	0.905	0.937	0.028	0.850	0.891	0.045	0.927	0.966	0.023
MDSAM (Gao et al., 2024)	0.919	0.950	0.025	0.881	0.913	0.040	0.941	0.970	0.020
SFEN (Du et al., 2025)	0.919	0.948	0.027	0.861	0.885	0.057	0.936	0.963	0.023
Ours	0.927	0.964	0.019	0.882	0.920	0.036	0.941	0.976	0.018
Methods	PASCAL-S			ECSSD					
	$S_\alpha \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	MAE $\downarrow$			
U2Net (Qin et al., 2020)	0.844	0.850	0.074	0.928	0.925	0.033			
ICON (Zhuge et al., 2022)	0.861	0.893	0.064	0.929	0.954	0.032			
EDN (Wu et al., 2022)	0.865	0.902	0.062	0.927	0.951	0.032			
MENet (Wang et al., 2023)	0.872	0.913	0.054	0.928	0.954	0.031			
MDSAM (Gao et al., 2024)	0.880	0.918	0.052	0.946	0.965	0.023			
SFEN (Du et al., 2025)	0.878	0.914	0.055	0.940	0.959	0.026			
Ours	0.892	0.936	0.041	0.950	0.975	0.018			

C. Qualitative Visualization

To provide a more intuitive understanding of our model’s superiority, we visualize the segmentation results across four specialized domains in Figures 5 to 8. As observed, our method consistently generates masks that are visually closest to the Ground Truth (GT). Crucially, the dynamic topology allows it to adapt to varying object scales and contrasts, preserving fine-grained details and reducing false positives where static PEFT baselines often fail.

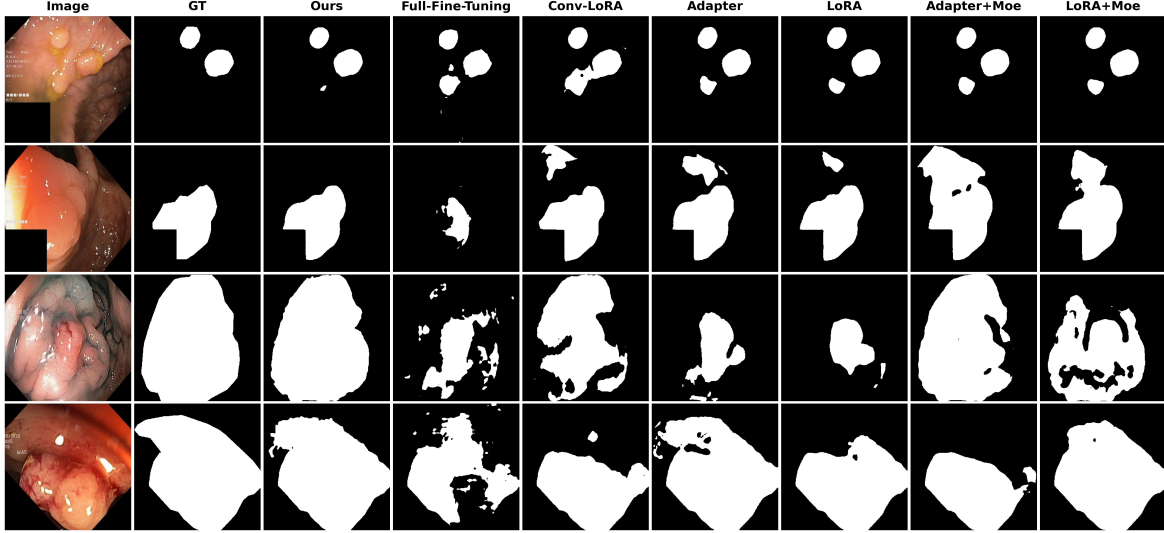


Figure 5. **Qualitative comparison on Polyp Segmentation (Kvasir-SEG).** Ours (3rd column) accurately delineates polyp boundaries even in low-contrast conditions, whereas baselines like Full-Fine-Tuning suffer from severe noise and fragmentation due to catastrophic forgetting.

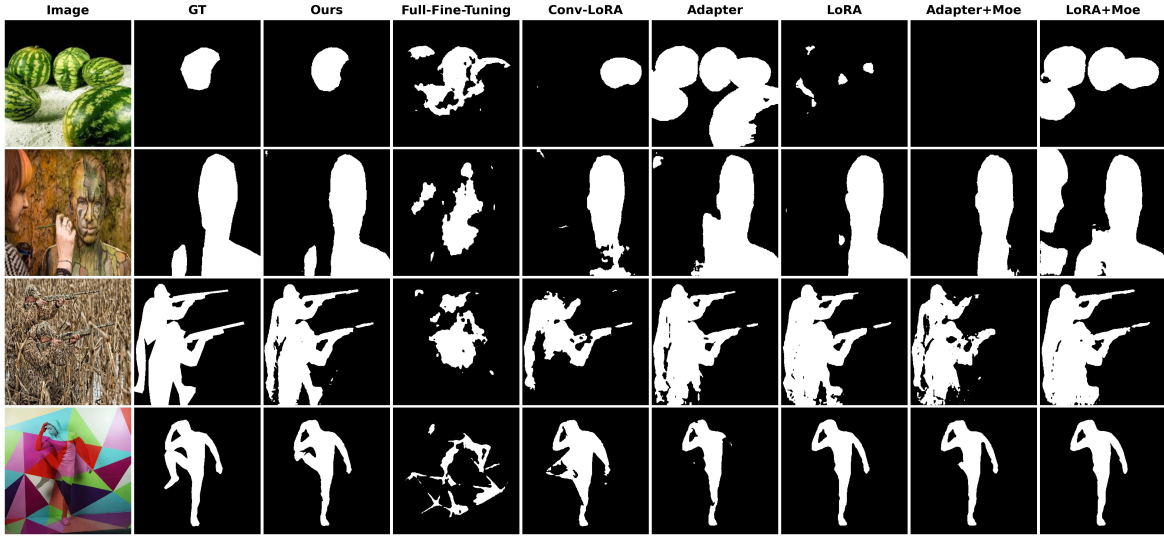


Figure 6. **Qualitative comparison on Camouflaged Object Detection (CAMO).** In scenarios with high background clutter (e.g., the soldier in the cornfield, 3rd row), our ACO-driven dynamic routing successfully distinguishes the target from the surroundings, achieving much sharper boundaries than competitors.

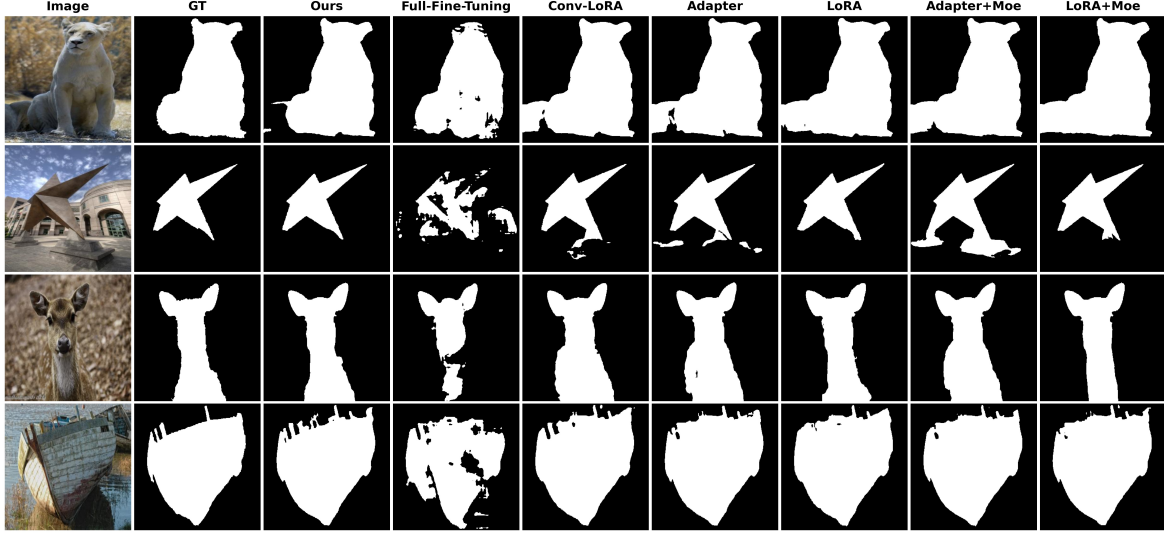


Figure 7. **Qualitative comparison on Salient Object Detection (ECSSD).** Our method demonstrates superior integrity in segmenting complex structures, avoiding the under-segmentation issues observed in static Adapter and LoRA methods.

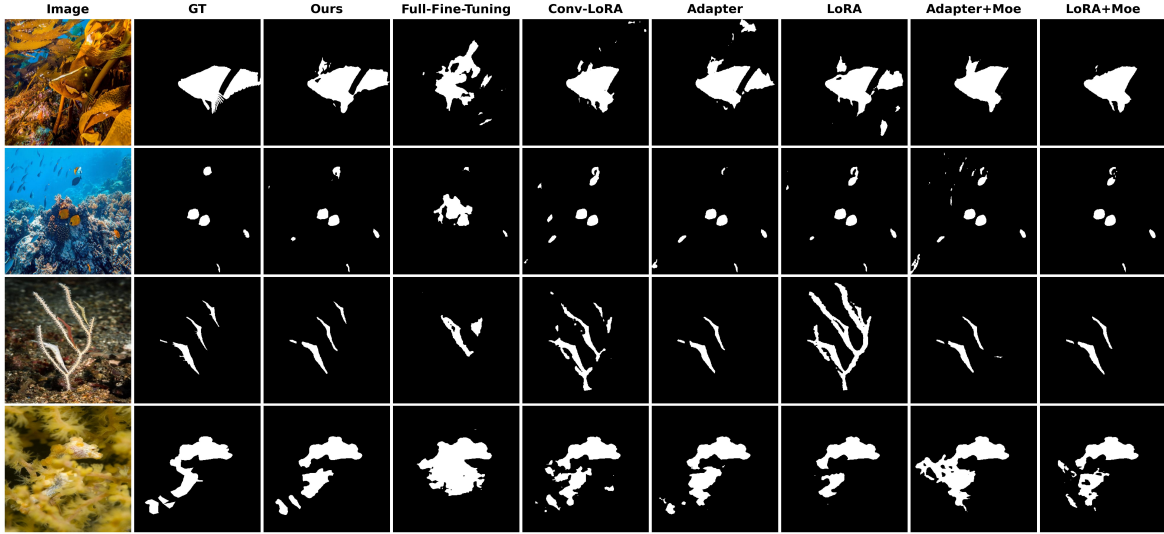


Figure 8. **Qualitative comparison on Marine Animal Segmentation (MAS3K).** Despite the challenges of water turbidity and low contrast, our model robustly segments marine life forms. Note the failure of Full-Fine-Tuning (4th column) to capture the complete object structure.

D. Additional Efficiency Comparisons

D.1. Computational Overhead

Table 9 presents the full computational overhead comparison, measured on a single V100 32GB GPU (batch size 16, 50 epochs). Our dynamic variant adds only $\sim 3\%$ training time and $\sim 12.5\%$ peak memory compared to standard LoRA, while achieving higher accuracy and lower FLOPs. Figure 9 plots the per-epoch training time for all three methods over 50 epochs.

Table 9. Computational overhead comparison.

Method	Param (M)	Train time (s)	Train mem (MB)	Test mem (MB)	GFlops
Standard LoRA	17.62	5874.10	17564	3217	134.41
Fixed-rank	3.73	5950.11	18410	2884	130.67
Dynamic	8.24	6065.53	19771	2967	129.76

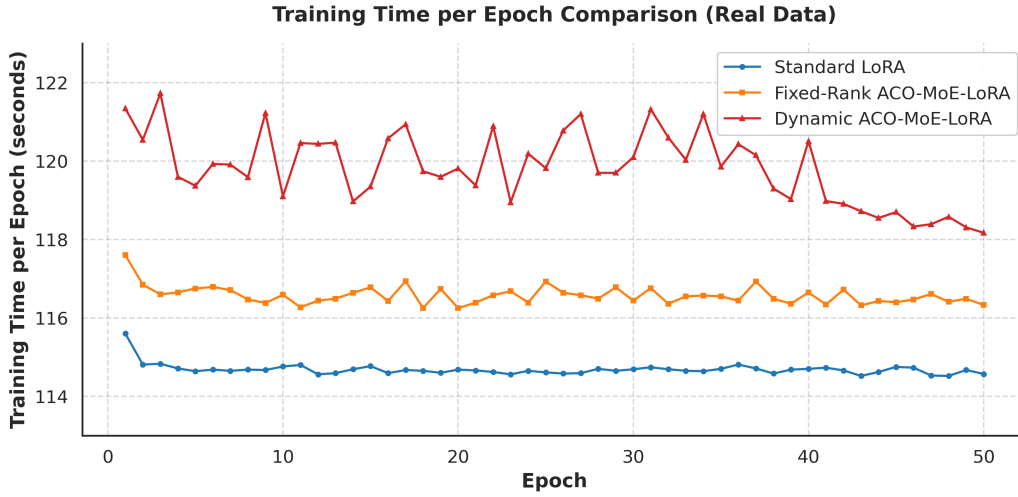


Figure 9. Per-epoch training time comparison.

D.2. Comparison with GA and PSO Meta-Controllers

To empirically justify the choice of ACO over GA and PSO, we implemented both meta-heuristics following standard practices. All methods share the same search space, batch size (16), and validation protocol. Training runs for 50 epochs on a V100 32GB GPU.

Quantitative Comparison Table 10 compares overall efficiency and final test performance. ACO achieves the highest mDice (0.871) with the shortest training time (1.68 h), while GA is nearly twice as slow (3.33 h) and performs worst. PSO offers similar training time to ACO but lower accuracy and higher FLOPs.

Table 10. Comparison of ACO, GA, and PSO as meta-controllers.

Method	Total Training Time	Peak GPU Mem (MB)	Test mDice	Test mIoU	GFlops
GA	3.33 h (11984 s)	20332	0.862	0.800	129.99
PSO	1.70 h (6119 s)	19766	0.868	0.805	134.02
ACO	1.68 h (6065 s)	19771	0.871	0.808	129.76

Stability Analysis We analyzed the validation IoU over the first 20 epochs (critical search phase). During the post-exploration stage (epochs 11–20), ACO exhibits the smallest standard deviation (0.0376), while GA (0.1732) and PSO (0.2823) show much larger fluctuations. For example, PSO drops to 0.132 at epoch 13 and GA repeatedly falls below 0.3.

To assess multi-seed robustness, we ran 5 independent seeds for 20 epochs (sufficient to observe convergence trends). Table 11 reports the results at epoch 20. ACO has the lowest run-to-run variability (std 0.004) and highest best IoU (0.871).

Table 11. Multi-seed stability (5 seeds, epoch 20).

Method	Mean Final IoU	Mean Best IoU	Mean Epoch to Best	Std Dev (5 seeds)
GA	0.693 ± 0.081	0.795 ± 0.035	15.8 ± 3.2	0.081
PSO	0.831 ± 0.015	0.846 ± 0.021	13.6 ± 5.1	0.015
ACO	0.841 ± 0.004	0.871 ± 0.003	14.2 ± 1.6	0.004

These results provide concrete objective evidence that ACO is better suited for our discrete combinatorial optimization task: it balances exploration and exploitation via pheromone feedback, avoids the inefficiency of GA and the instability of PSO, and its online pheromone update naturally fits our “Evolving-while-Training” paradigm.

E. Hyperparameter Sensitivity Analysis

E.1. Sensitivity to ACO Hyperparameters (M and ρ)

We analyze the sensitivity of two key ACO hyperparameters: the number of ants M and the evaporation rate ρ . Experiments are conducted on five Polyp segmentation datasets (Kvasir-SEG, CVC-ClinicDB, CVC-ColonDB, CVC-300, ETIS). Table 12 reports mDice and mIoU for different (M, ρ) configurations.

Table 12. Full hyperparameter sensitivity analysis (Polyp benchmark).

(M, ρ)	Kvasir-SEG		CVC-ClinicDB		CVC-ColonDB		CVC-300		ETIS	
	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
(10, 0.3)	0.930	0.883	0.900	0.845	0.802	0.727	0.892	0.825	0.818	0.739
(30, 0.3)	0.928	0.880	0.918	0.864	0.811	0.737	0.889	0.823	0.799	0.728
(40, 0.3)	0.931	0.883	0.909	0.853	0.796	0.720	0.885	0.819	0.803	0.729
(20, 0.1)	0.935	0.886	0.916	0.865	0.806	0.728	0.886	0.820	0.805	0.729
(20, 0.5)	0.935	0.890	0.913	0.854	0.804	0.725	0.882	0.815	0.807	0.731
(20, 0.7)	0.923	0.872	0.899	0.839	0.812	0.731	0.898	0.830	0.804	0.726
(20, 0.3) (default)	0.933	0.887	0.914	0.864	0.814	0.740	0.890	0.825	0.804	0.725

The results show that the method is stable across a wide range of M (10–40) and ρ (0.1–0.5), with only a slight performance decrease at the extremes ($M = 40$ or $\rho = 0.7$). The default configuration ($M = 20$, $\rho = 0.3$) achieves the best overall performance and is recommended for new tasks.

E.2. Ablation of Scaling Hyperparameter α

To justify the necessity of optimizing α in our dynamic setting, we compare fixed α values with our ACO-optimized α . Table 13 presents the results on the same five Polyp datasets.

Table 13. Ablation study of scaling hyperparameter α (Polyp benchmark).

α	Kvasir-SEG		CVC-ClinicDB		CVC-ColonDB		CVC-300		ETIS	
	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
$\alpha = 4$	0.933	0.887	0.905	0.857	0.808	0.733	0.869	0.805	0.785	0.713
$\alpha = 8$	0.924	0.879	0.906	0.856	0.792	0.719	0.883	0.817	0.810	0.737
$\alpha = 16$	0.924	0.874	0.903	0.855	0.797	0.721	0.884	0.814	0.788	0.709
$\alpha = 32$	0.930	0.881	0.889	0.841	0.785	0.710	0.891	0.822	0.788	0.710
Dynamic (ACO-opt.)	0.933	0.887	0.914	0.864	0.814	0.740	0.890	0.825	0.804	0.725

The ACO-optimized α consistently outperforms all fixed α values, achieving the highest average mDice and mIoU. This confirms that dynamic α tuning is beneficial in our evolving-rank setting.